AI-AUGMENTED BACKEND DEVELOPMENT

AI BLOGNEST API

PHASE 3 – PROJECT DESIGN PHASE

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Submitted as a phase-wise project submission.

1. Introduction

The Project design phase defines how the AI BlogNest API is organized. The system follows a modular backend structure so that requests, business logic, database operations and AI services can be handled separately.

2. Architecture Components

- API Interface: Receives client requests and returns JSON responses.
- Express Server Gateway: Handles HTTP methods, routes, JSON data and CORS.
- Authentication Middleware: Validates tokens and user access.
- Service Layer: Contains business logic for blogs, comments and other operations.
- Database Layer: Uses Mongoose to communicate with MongoDB.
- AI Flow: Connects AI services for blog generation and summarization.

3. MVC Pattern

- Model: Defines data structures and Mongoose schemas.
- Controller: Processes requests, applies logic and prepares responses.
- Routing/API Layer: Maps API endpoints to controller operations.

4. Database Relationships

- One user can create multiple posts.
- One user can create multiple comments.
- One post can contain multiple comments.
- Post and user information are connected through references.

5. Security Architecture

- JWT-based authentication.
- Role-based route protection.
- Password hashing.
- Input validation and sanitization.
- Centralized error handling.

6. Expected Outcome

A clear layered architecture is prepared so that the backend remains modular, secure and easier to maintain.

7. Conclusion

The design phase provides the structural foundation for implementing the AI BlogNest API and integrating its database, security and AI services.